

Bob's Donut Cafe | Introduction to the one-sample t-test



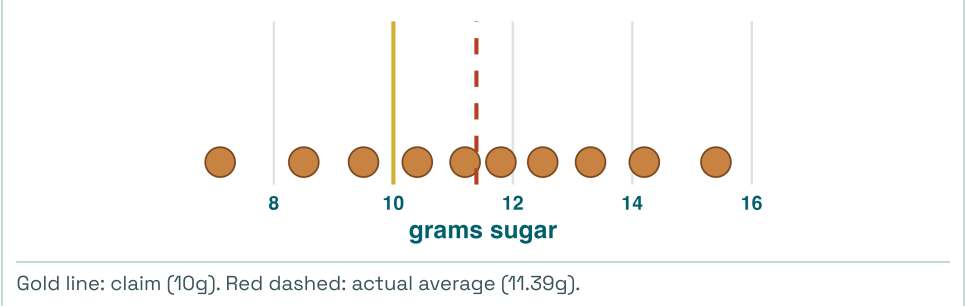
The Plan

- Buy a donut on ten random days.
- Lab-test sugar.

The Results

```
sugar <- c(7.1, 8.5, 9.5, 10.4, 11.2, 11.8, 12.5, 13.3, 14.2, 15.4)
mean(sugar)
#> 11.39
```

```
ggplot(tibble(sugar), aes(sugar)) +
  geom_dotplot(binwidth = .5, colour = "#7a4a22") +
  geom_vline(xintercept = 10, colour = "#d4af33") +
  geom_vline(xintercept = mean(sugar), colour = "#b5451f",
    linetype = "dashed")
```



hmmmm... is this enough evidence that Bob is cheating?

Two possibilities:

- Recipe really is 10g, 11.39g is just natural variation.
- Recipe really is above 10g.

Which one? Let's settle it with a formal hypothesis test.

Hypothesis test framework

State a **null hypothesis** (H_0) and an **alternative hypothesis** (H_1).

H_0 is the status quo. The true mean sugar is 10g. $H_0 : \mu = 10$

H_1 is what we want evidence for. Two flavors:

ONE-SIDED (WE USE FIRST)

$H_1 : \mu > 10$

true mean sugar exceeds the claim.

TWO-SIDED (THE DEFAULT)

$H_1 : \mu \neq 10$

true mean sugar differs from the claim in either direction.

Quick check: name the hypotheses

A drug company claims its pill lowers blood pressure. What is H_0 ? ✓ B

A. The pill lowers blood pressure

B. The pill has no effect

C. The pill raises blood pressure

A factory's bolts should be 5cm. You suspect the machine has drifted out of calibration and want to test this. Which H_1 ? ✓ C

A. $\mu > 5$

B. $\mu < 5$

C. $\mu \neq 5$

Measuring the gap: Is 1.39g a lot?

By itself, **1.39g is unclear**. We need another scale, one that accounts for how much the sugar content naturally varies among donuts, and how many donuts we weighed.

Variance: how scattered?

High variance

less surprising

Low variance

surprising

Sample size: how much data?

2 donuts

1.39g diff doesn't mean much

100s of donuts

might point to real difference

For the same average gap, the evidence for a significant difference is stronger when the data are less variable / consistent and the sample size is larger / smaller. ✓ variable • larger

Step 1: standard deviation, s

s measures how much individual donuts vary. Average distance from the mean.

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n - 1}} = 2.59 \text{ g}$$

Step 2: standard error, SE

Means wobble less than single donuts. Divide s by $\sqrt{n}$.

$$SE = \frac{s}{\sqrt{n}} = \frac{2.59}{\sqrt{10}} = 0.82 \text{ g}$$

Step 3: t statistic

Standardised gap: how many SEs above the claim?

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{1.39}{0.82} = 1.70$$

```
s <- sd(sugar)
se <- s / sqrt(length(sugar))
t <- (mean(sugar) - 10) / se
#> t = 1.70
```

Step 4: p -value

Compare t to a t curve with $df = n - 1 = 9$. One-sided $H_1 : \mu > 10$, so p = the upper tail only.

$t = 1.70$, upper-tail $p = 0.062$.

```
pt(t, df = 9, lower.tail = FALSE)
#> 0.062
```

All-in-one in R

```
t.test(sugar, mu = 10, alternative = "greater")
```

One Sample t-test

data: sugar
t = 1.6972, df = 9, p-value = 0.06195
alternative hypothesis: true mean is greater than 10
#> ... (CI + estimates clipped) ...

One-sided $p = 0.062 > 0.05$: suggestive, but not quite enough to reject H_0 at $n = 10$.

Practice reading one-sample t-tests, then extend Bob's audit

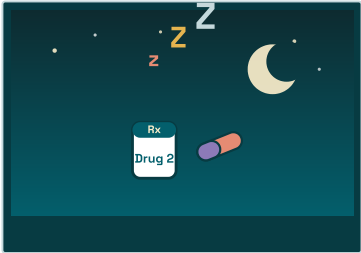
The built-in datasets::sleep data record extra hours of sleep after two drugs. Here we test whether drug 2 adds more than 0 hours on average.

```
data(sleep, package = "datasets")
drug2 <- subset(sleep, group == 2)$extra
drug2
# [1] 1.9 0.8 1.1 0.1 -0.1 4.4 5.5
1.6 4.6 3.4
```

```
t.test(drug2, mu = 0,
       alternative = "greater")
```

One Sample t-test
data: drug2
t = 3.6799, df = 9, p-value = 0.002538
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
1.169334 Inf
sample estimates:
mean of x
2.33

R



1. What is the null hypothesis being tested?
✓ A

A. $H_0 : \mu = 0$
B. $H_0 : \mu > 0$
C. $H_0 : \mu = 2.33$

2. What is the sample mean extra sleep? ✓ B

A. 0
B. 2.33
C. 3.6799

3. How many observations were tested? ✓ C

A. 9
B. 2
C. 10

4. Match each term to its value, then to its plain-English meaning. Some values are rounded.

Values: a. 9 · b. 0.002538 · c. [1.17, ∞) · d. 3.68

TERM	VALUE	MEANING
t	d. 3.68	b. ≈ # of SDs sample mean is above 0
df	a. 9	a. sample size minus one (n - 1)
p-value	b. 0.002538	d. P(t this extreme H_0 true)
95% CI	c. [1.17, ∞)	c. plausible range for the true mean μ

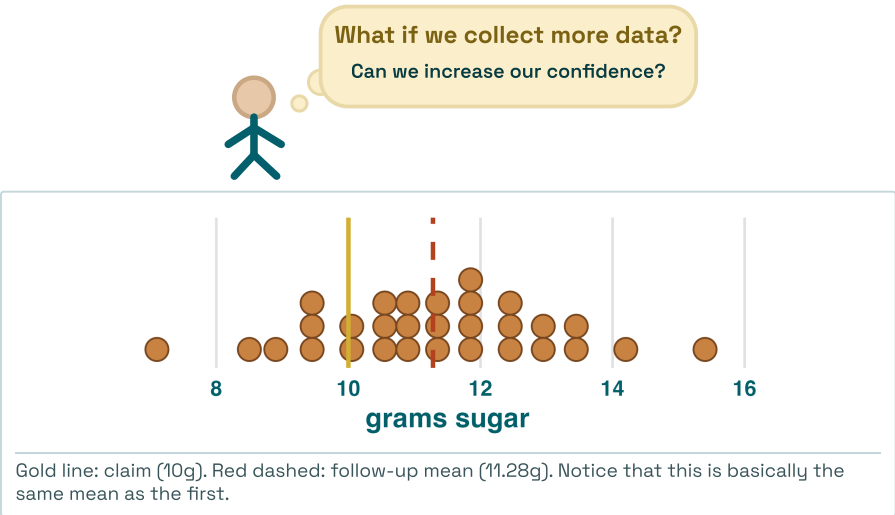
Meanings: a. n - 1 · b. ≈ # of SDs the sample mean sits above 0 · c. plausible range for μ · d. P(t this extreme | H_0)

5. At $\alpha = 0.05$, what is the decision? ✓ B

A. Fail to reject H_0
B. Reject H_0
C. H_0 is proven true

Back to the audit of Bob's donuts: More data

The first donut audit had $\bar{x} = 11.39$ and one-sided $p = 0.062$. Close, but not below $\alpha = 0.05$.



```
sugar30 <- c(7.1, 8.5, 9.5, 10.4, 11.2,
11.8, 12.5, 13.3, 14.2, 15.4,
10.9, 9.6, 11.7, 12.3, 10.2,
11.0, 13.1, 9.9, 12.6, 10.7,
11.5, 10.4, 12.0, 9.3, 11.9,
13.6, 10.8, 11.3, 12.8, 8.9)
```

R

```
t.test(sugar30, mu = 10,
       alternative = "greater")
```

R

One Sample t-test
data: sugar30
t = 3.9266, df = 29, p-value = 0.000244
alternative hypothesis: true mean is greater than 10
95 percent confidence interval:
10.72612 Inf
sample estimates:
mean of x
11.28

Manual calculations

$s = 1.79 \text{ g}$

$SE = \frac{s}{\sqrt{n}} = \frac{1.79}{\sqrt{30}} = 0.33$

$t = \frac{11.28 - 10}{0.33} = 3.93$

```
s <- sd(sugar30)
#> 1.79
```

```
se <- s / sqrt(30)
#> 0.33
```

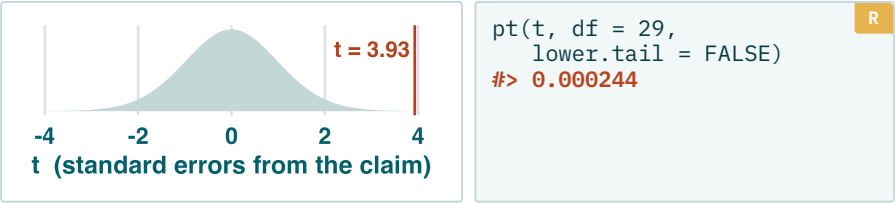
```
t <- (mean(sugar30) - 10) / se
#> 3.93
```

R

R

R

Convert t to p



Interpret the follow-up test

6. What stayed the same from the first donut test? ✓ A

A. $H_0 : \mu = 10$ and $H_1 : \mu > 10$
B. The sample size
C. The p-value

7. Why is the evidence now stronger? ✓ C

A. The claim changed
B. The sample mean became significantly larger
C. More data made the SE smaller

8. At $\alpha = 0.05$, what is the decision now? ✓ B

A. Fail to reject H_0
B. Reject H_0
C. Prove H_0 false

9. In context, the follow-up audit gives strong evidence that Bob's true mean added sugar is ____ 10g. (above / below / equal to) ✓ above

10. Which interpretation is most careful? ✓ C

A. Every donut has more than 10g of sugar
B. Bob is probably adding 11.28g of sugar to all his donuts
C. The data are unlikely if Bob is truly adding 10g sugar on average